

Node.js and Express Foundations

A Simple Node.js and Express Server



Jon Friskics

Senior Software Development Author, Pluralsight

@jonfriskics@mastodon.social | www.pluralsight.com



Server



Client



Server



Client



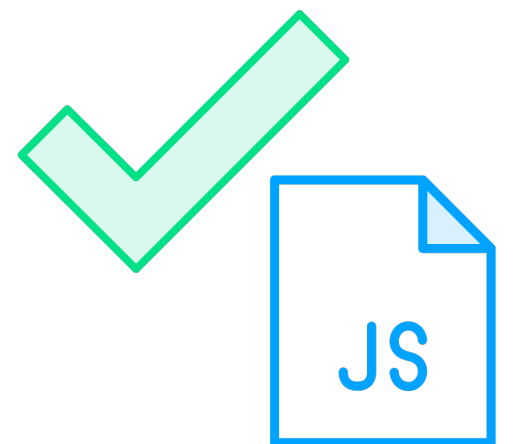
request

`http://server/path`

Server



Client



response



request

`http://server/path`

Server

(running Node.js)



Client

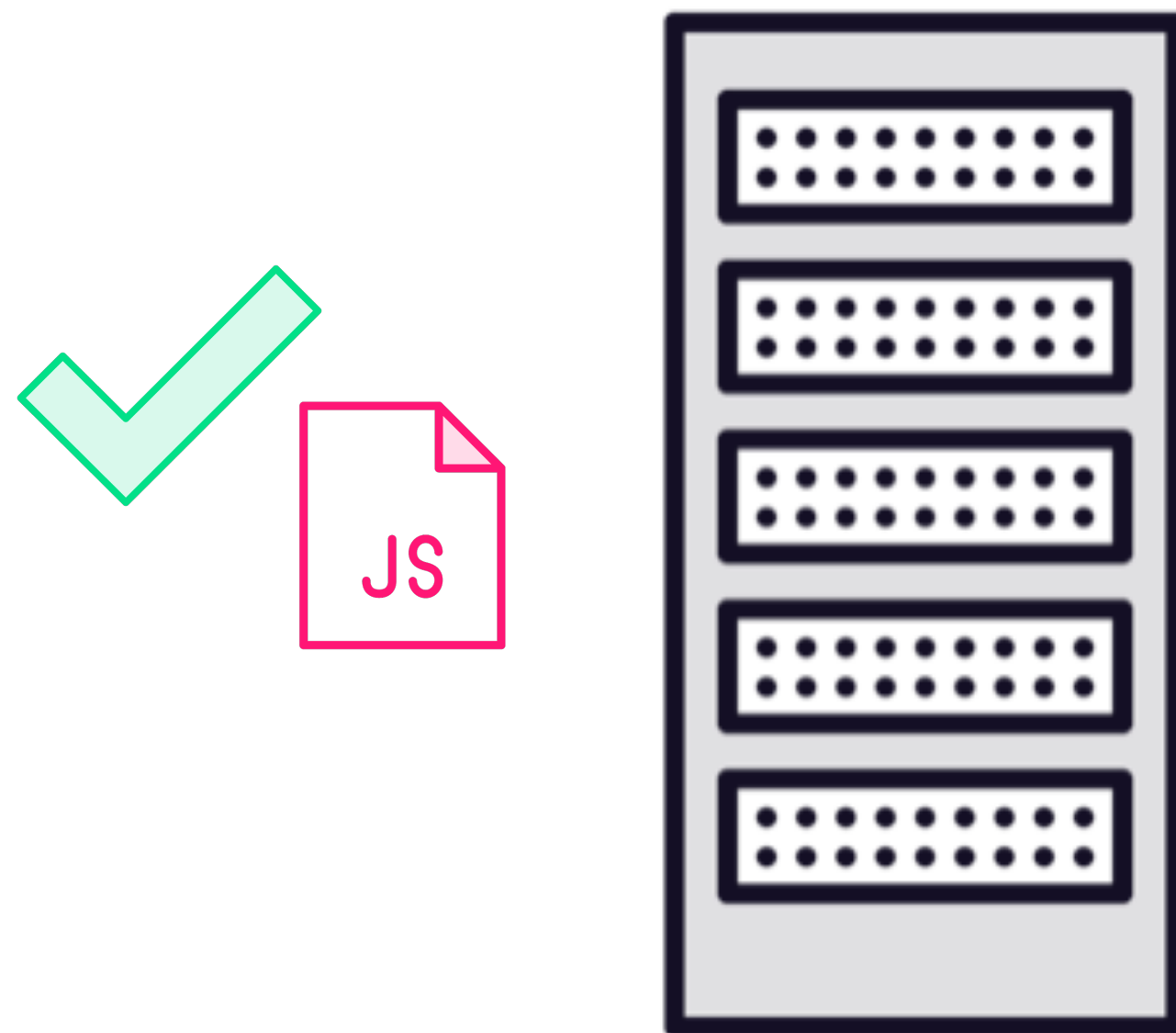


request

`http://server/path`

Server

(running Node.js)



response

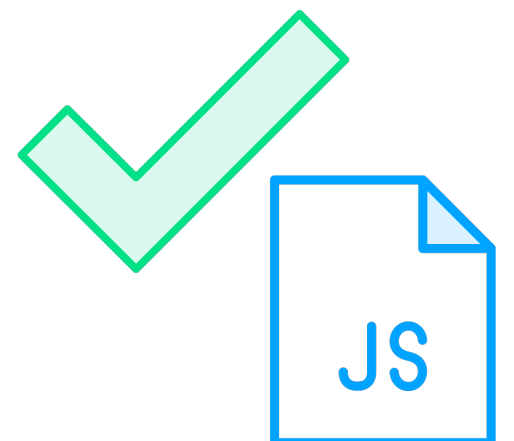


request

http://server/path

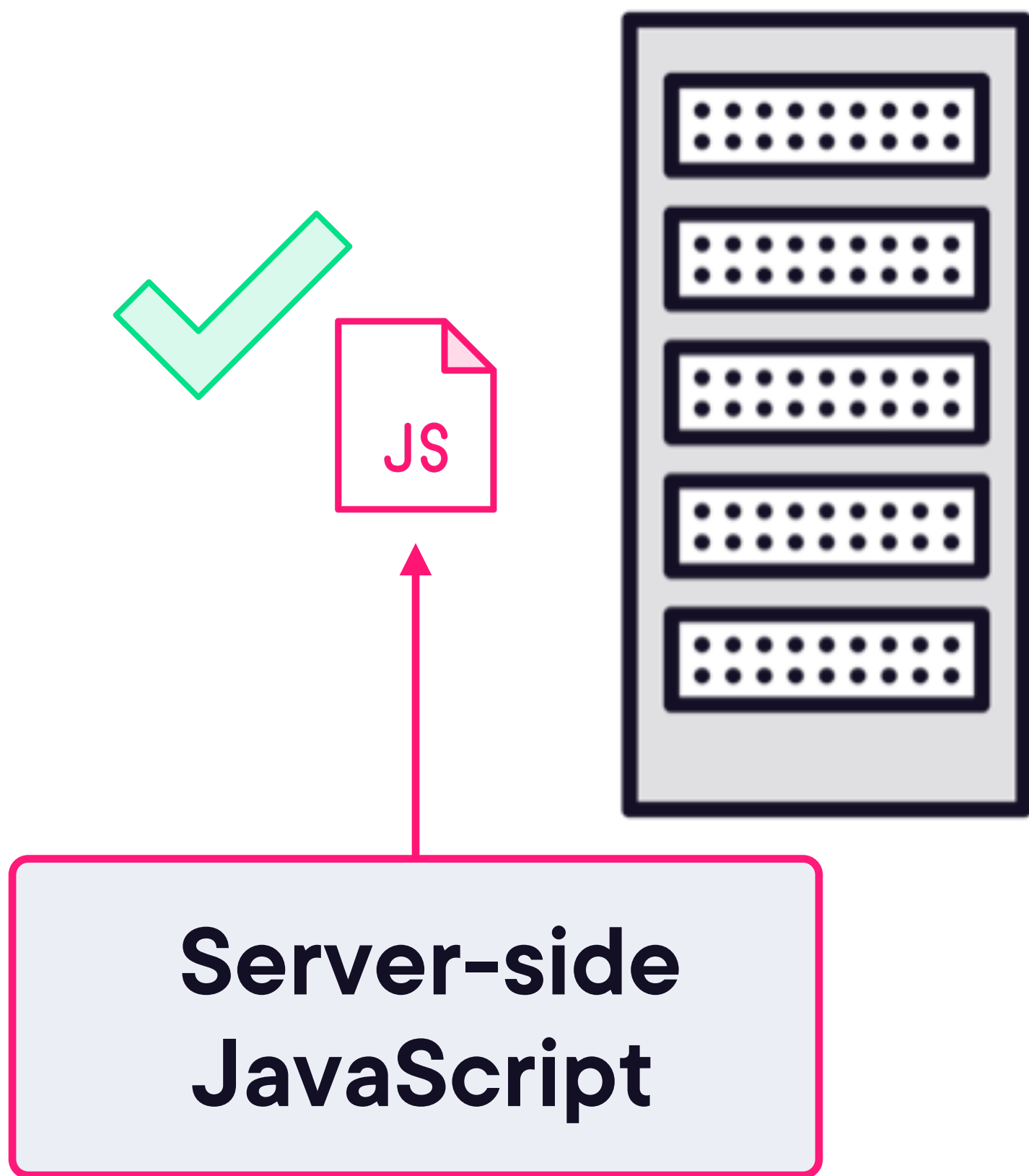


Client

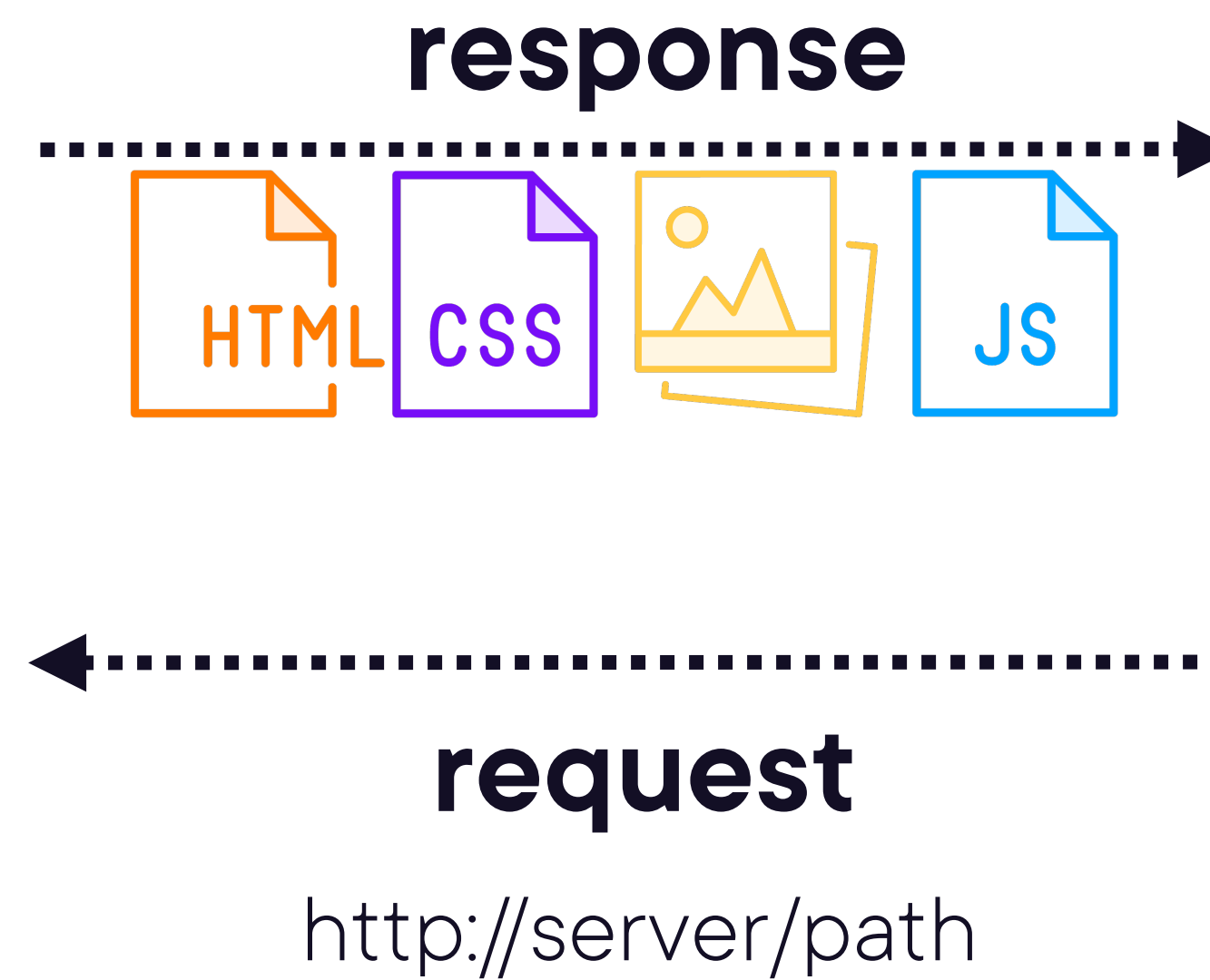


Server

(running Node.js)



Client



A Simple Node Server

app.mjs

```
import { createServer } from 'node:http'

const server = createServer()

function requestCallback(request, response) {
  response.writeHead(200)
  response.end(`request received`)
}

server.on('request', requestCallback)

server.listen(3000)
```

A Simple Node Server

app.mjs

```
import { createServer } from 'node:http'

const server = createServer()

function requestCallback(request, response) {
  response.writeHead(200)
  response.end(`request received`)
}

server.on('request', requestCallback)

server.listen(3000)
```

Server



A Simple Node Server

app.mjs

```
import { createServer } from 'node:http'

const server = createServer()

function requestCallback(request, response) {
  response.writeHead(200)
  response.end(`request received`)
}

server.on('request', requestCallback)

server.listen(3000)
```

Server



Client



request

http://localhost:3000

A Simple Node Server

app.mjs

```
import { createServer } from 'node:http'

const server = createServer()

function requestCallback(request, response) {
  response.writeHead(200)
  response.end(`request received`)
}

server.on('request', requestCallback)

server.listen(3000)
```

Server



Client



request

http://localhost:3000

A Simple Node Server

app.mjs

```
import { createServer } from 'node:http'

const server = createServer()

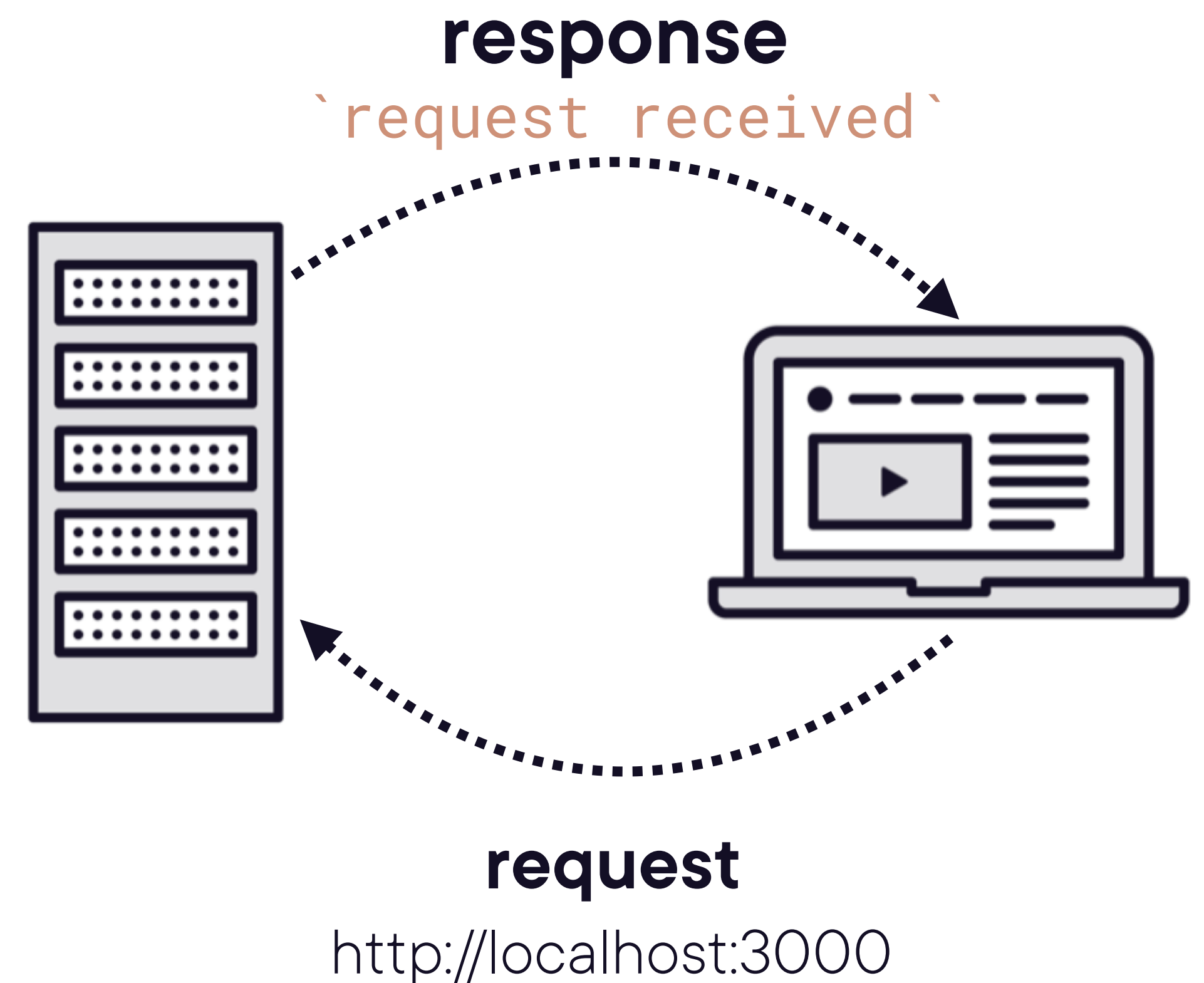
function requestCallback(request, response) {
  response.writeHead(200)
  response.end(`request received`)
}

server.on('request', requestCallback)

server.listen(3000)
```

Server

Client



Node.js Server Creation Pattern

Three main actions to create and use a server

1

Import some code and create a server object

2

Listen for requests coming in to that server

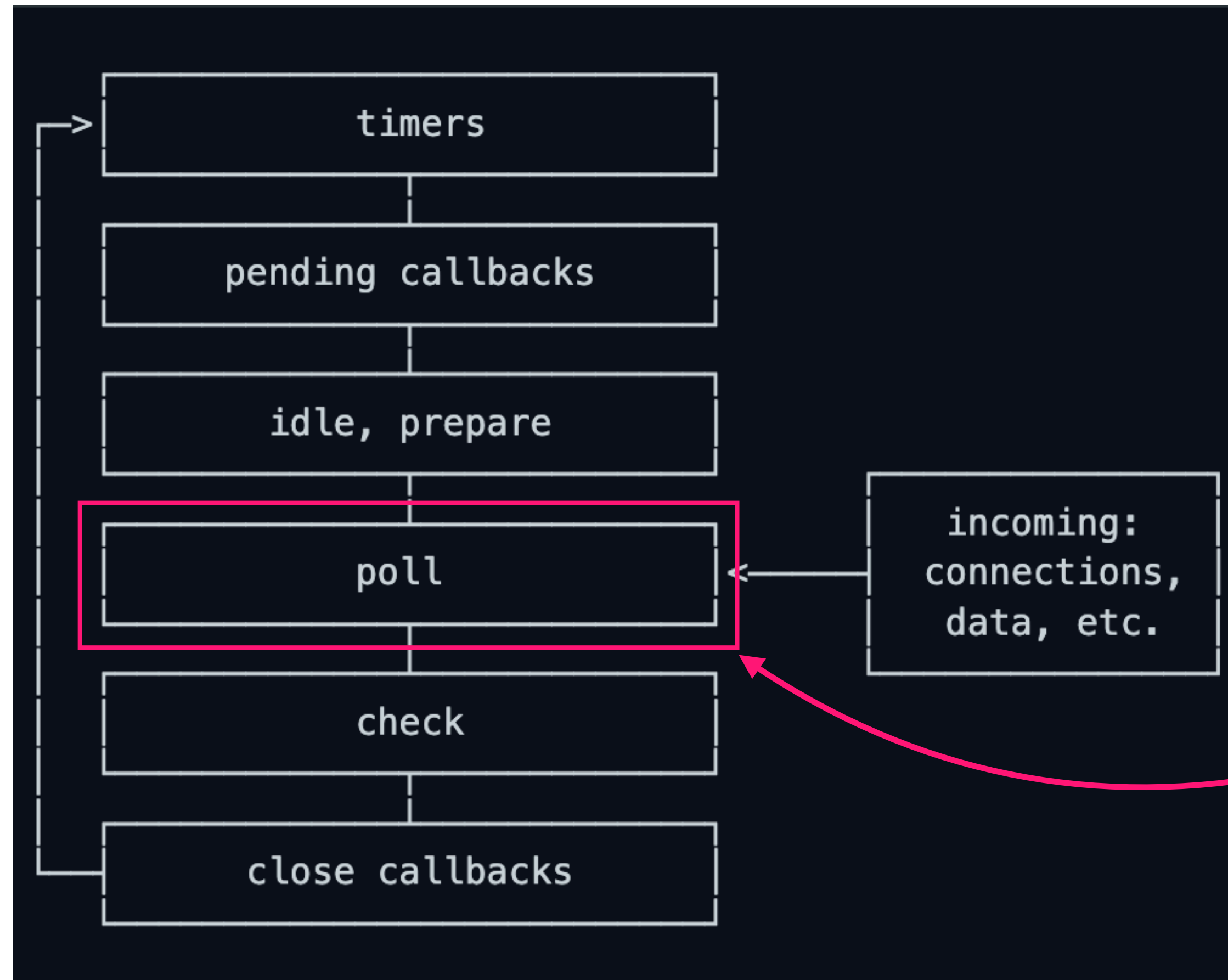
3

Define callback functions that run when that request event happens

Node.js Event Loop

Node.js programs loop forever until they are stopped

Node.js Event Loop



**server requests and
client responses happen
in this phase**

A Simple Express Server

app-express.mjs

```
import express from 'express'

const server = express()

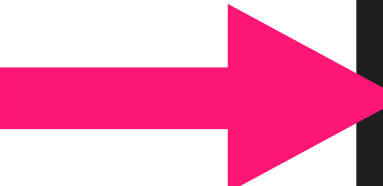
function requestCallback(request, response) {
  response.status(200).send(`request received`)
}

server.get('/', requestCallback)

server.listen(3000)
```

A Simple Express Server

app-express.mjs



```
import express from 'express'

const server = express()

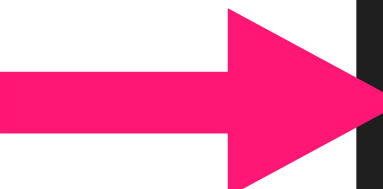
function requestCallback(request, response) {
  response.status(200).send(`request received`)
}

server.get('/', requestCallback)

server.listen(3000)
```

A Simple Express Server

app-express.mjs



```
import express from 'express'

const server = express()

function requestCallback(request, response) {
  response.status(200).send(`request received`)
}

server.get('/', requestCallback)

server.listen(3000)
```

A Simple Express Server

app-express.mjs

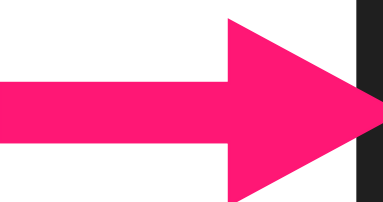
```
import express from 'express'

const server = express()

function requestCallback(request, response) {
  response.status(200).send(`request received`)
}

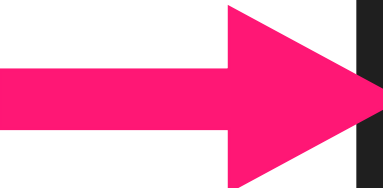
server.get('/', requestCallback)

server.listen(3000)
```



A Simple Express Server

app-express.mjs



```
import express from 'express'

const server = express()

function requestCallback(request, response) {
  response.status(200).send(`request received`)
}

server.get('/', requestCallback)

server.listen(3000)
```

A Simple Express Server

app-express.mjs

```
import express from 'express'

const server = express()

function requestCallback(request, response) {
  response.status(200).send(`request received`)
}

server.get('/', requestCallback)

server.listen(3000)
```

Server



A Simple Express Server

app-express.mjs

```
import express from 'express'

const server = express()

function requestCallback(request, response) {
  response.status(200).send(`request received`)
}

server.get('/', requestCallback)

server.listen(3000)
```

Server



A Simple Express Server

app-express.mjs

```
import express from 'express'

const server = express()

function requestCallback(request, response) {
  response.status(200).send(`request received`)
}

server.get('/', requestCallback)

server.listen(3000)
```

Server



A Simple Express Server

app-express.mjs

```
import express from 'express'

const server = express()

function requestCallback(request, response) {
  response.status(200).send(`request received`)
}

server.get('/', requestCallback)

server.listen(3000)
```

Server



Client



GET request

`http://localhost:3000`

A Simple Express Server

app-express.mjs

```
import express from 'express'

const server = express()

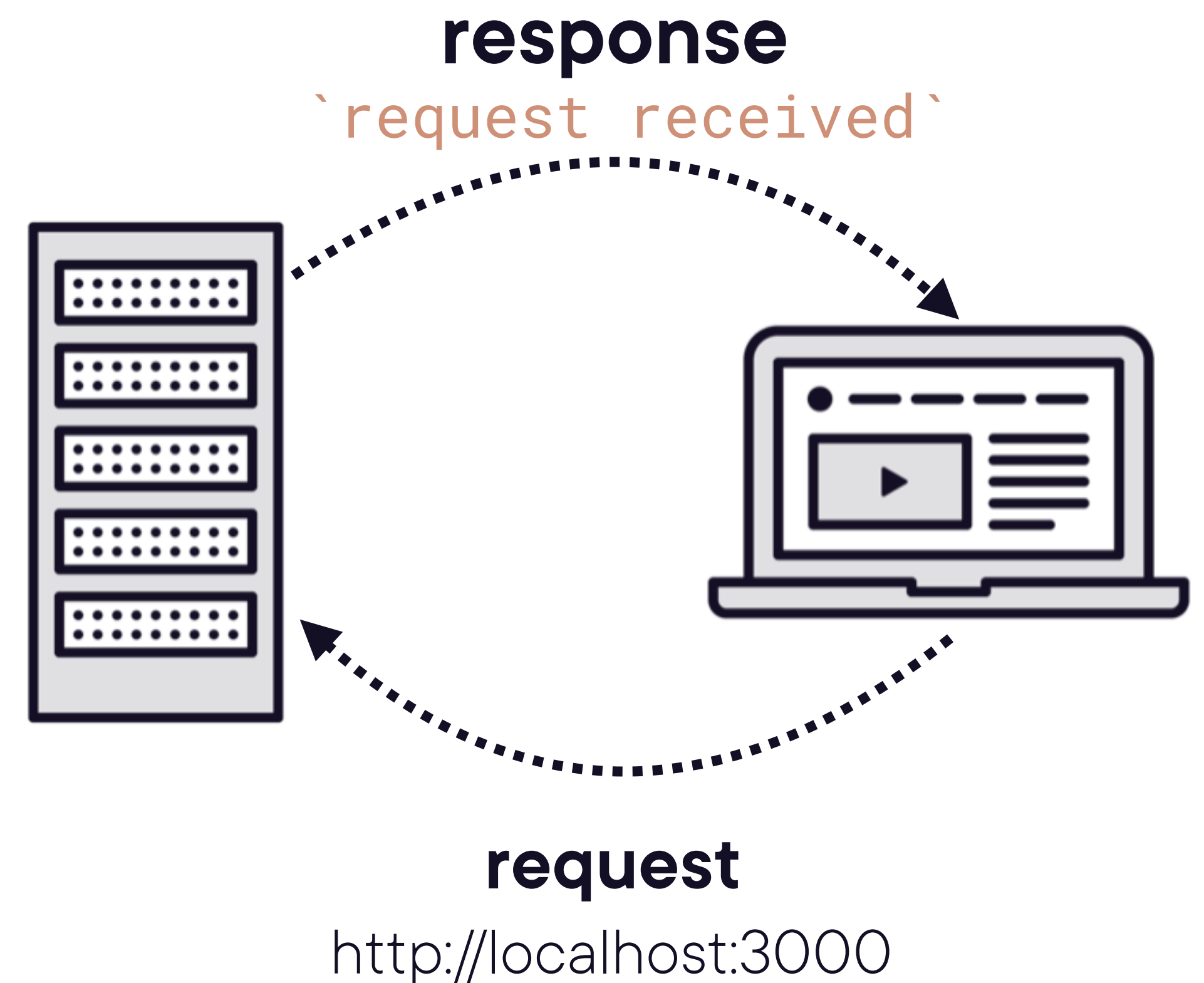
function requestCallback(request, response) {
  response.status(200).send(`request received`)
}

server.get('/', requestCallback)

server.listen(3000)
```

Server

Client



HTTP Status Code 200

"OK"

The request succeeded